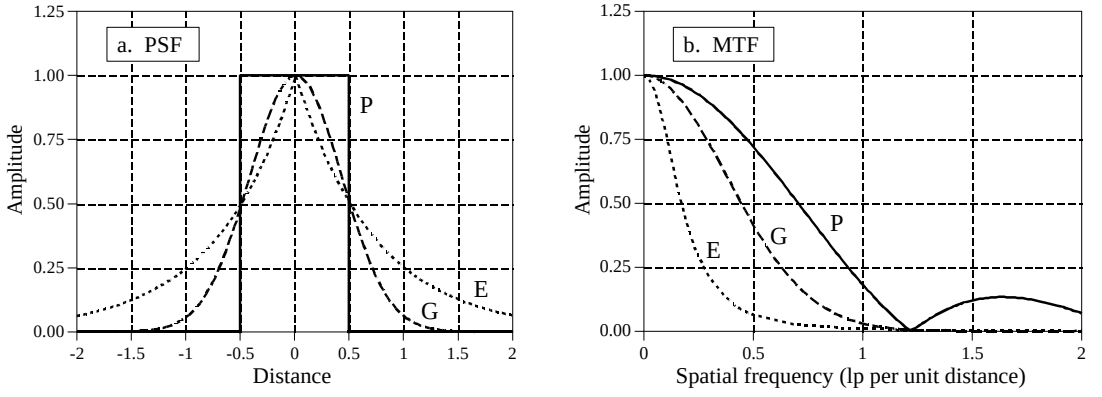

FIGURE 25-1
FWHM versus MTF. Figure (a) shows profiles of three PSFs commonly found in imaging systems: (P) pillbox, (G) Gaussian, and (E) exponential. Each of these has a FWHM of one unit. The corresponding MTFs are shown in (b). Unfortunately, similar values of FWHM do not correspond to similar MTF curves.

light. This is used in radiation detectors, night vision light amplifiers, and CRT displays. The exact shape of these three PSFs is not important for this discussion, only that they broadly represent the PSFs seen in real world applications.

The PSF contains complete information about the spatial resolution. To express the spatial resolution by a single number, we can ignore the *shape* of the PSF and simply measure its *width*. The most common way to specify this is by the Full-Width-at-Half-Maximum (FWHM) value. For example, all the PSFs in (a) have an FWHM of 1 unit.

Unfortunately, this method has two significant drawbacks. First, it does not match other measures of spatial resolution, including the subjective judgement of observers viewing the images. Second, it is usually very difficult to directly measure the PSF. Imagine feeding an impulse into an imaging system; that is, taking an image of a very small white dot on a black background. By definition, the acquired image will be the PSF of the system. The problem is, the measured PSF will only contain a few pixels, and its contrast will be low. Unless you are very careful, random noise will swamp the measurement. For instance, imagine that the impulse image is a 512×512 array of all zeros except for a single pixel having a value of 255. Now compare this to a normal image where all of the 512×512 pixels have an average value of about 128. In loose terms, the signal in the impulse image is about 100,000 times weaker than a normal image. No wonder the signal-to-noise ratio will be bad; there's hardly any signal!

A basic theme throughout this book is that signals should be understood in the domain where the information is encoded. For instance, audio signals should be dealt with in the frequency domain, while image signals should be handled in the spatial domain. In spite of this, one way to measure image resolution is by looking at the *frequency response*. This goes against the fundamental